MDS211 Nervous System — SA1 Mock Exam 2

Lectures 1–16 · 77 Questions · Single Best Answer

This is a SECOND, fully independent 77-question set — zero overlap with Mock Exam 1 — using the same real SA1 weighting from your official schedule (proportional to actual lecture hours, 15 hours total across Lec 1–16). Every question again comes from my verified, fact-checked bank (the same one behind "The Professor's Gauntlet"), not the error-prone original archive. Use this as a second timed practice run once you've reviewed Mock Exam 1's answer key — budget roughly 2 minutes per question, matching the real SA1's pacing.

Lecture 1 — Introduction & Organization of the CNS (5 questions)

1. Which statement about gray and white matter distribution is correct?

A. In the spinal cord, gray matter is peripheral and white matter is central
B. In the cerebrum, gray matter forms the cortex and white matter lies deep to it
C. The cerebellum has no gray matter
D. White matter contains neuron cell bodies while gray matter contains only axons
E. Gray and white matter are distributed identically in brain and spinal cord

† original question, written from lecture content

2. A newborn is found to have a portion of spinal cord and meninges protruding through a vertebral defect, covered only by a thin membrane. This is best classified as:

A. Spina bifida occulta
B. Meningocele
C. Myelomeningocele
D. Anencephaly
E. Encephalocele

† original question, written from lecture content

3. Which cranial nerve numbers correspond to nerves that are purely sensory?

A. III, IV, VI
B. I, II, VIII
C. V, VII, X
D. XI, XII
E. III, VII, X

† original question, written from lecture content

4. Dorsal root ganglia contain the cell bodies of which type of neuron?

A. Lower motor neurons
B. Preganglionic autonomic neurons
C. First-order primary sensory neurons
D. Interneurons of the dorsal horn
E. Postganglionic sympathetic neurons

† original question, written from lecture content

5. A patient has a lesion that damages both anterior and posterior spinal nerve roots equally at one level. Which functional deficit pattern is expected?

A. Pure sensory loss only
B. Pure motor loss only
C. Combined sensory and motor loss
D. No deficit, roots regenerate fast
E. Only autonomic dysfunction occurs

† original question, written from lecture content

Lecture 2 — Neurohistology (Neuron & Neuroglia) (3 questions)

6. If the tract carrying FAST, sharp, well-localized pain is damaged, which fiber type is responsible for that lost sensation?

A. A-β fiber pathway
B. C fiber, unmyelinated pathway
C. Paleospinothalamic tract fibers
D. Neo-spinothalamic tract, A-δ fibers
E. A-α fiber pathway

7. Which of the following is NOT a component of the blood-brain barrier?

A. Oligodendrocyte
B. Astrocyte end-feet
C. Pericyte of capillary
D. Endothelial cell tight junctions
E. Basement membrane

8. Which cell type is the resident immune cell of the CNS, analogous to macrophages elsewhere in the body?

A. Microglia
B. Schwann cell
C. Oligodendrocyte
D. Astrocyte
E. Ependymal cell

Lecture 3 — Properties of Neuron: RMP, Action Potential, Conduction (5 questions)

9. Which of the following correctly describes when Na^+/K^+-ATPase activity increases the most?

A. During high neuronal firing, restoring gradients
B. Only during sleep states
C. Only in glial cells, never neurons
D. Constant rate, unchanging with activity
E. Only during hyperpolarization phases

10. When depolarizations from multiple simultaneous synaptic inputs on different parts of a neuron add together, this is called:

A. Temporal summation
B. Spatial summation
C. Facilitation
D. Long-term potentiation
E. Accommodation

11. All of the following are true of chemical synapses EXCEPT:

A. They involve a synaptic delay of roughly 0.5-1 ms
B. Transmission is unidirectional (pre- to postsynaptic)
C. They can amplify a small presynaptic signal into a larger postsynaptic response
D. Transmission is instantaneous with essentially zero delay
E. Neurotransmitter release is Ca^{2+}-dependent

12. Which statement about graded potentials is INCORRECT?

A. They decay in amplitude with distance (decremental conduction)
B. Their amplitude is proportional to stimulus strength
C. They can summate
D. They obey the all-or-none law like action potentials
E. They are typically local, subthreshold events

13. During the absolute refractory period, why can no new action potential be generated no matter how strong the stimulus?

A. All voltage-gated K^+ channels are closed
B. Voltage-gated Na^+ channels are inactivated and cannot reopen until the membrane repolarizes
C. The membrane is hyperpolarized below threshold, requiring only a slightly stronger stimulus
D. Cl^- channels are maximally open
E. The neuron has run out of neurotransmitter

† original question, written from lecture content

Lecture 4 — Metabolic Pathways of Neurons & Nervous Tissue (5 questions)

14. Why can hyperammonemia impair neuronal function even though ammonia detoxification happens mainly in astrocytes rather than neurons?

A. Astrocyte dysfunction disrupts neuronal fuel/NT recycling
B. Ammonia only affects neurons, not astrocytes
C. Neurons have glutamine synthetase too
D. Ammonia directly destroys myelin sheaths
E. No connection between the two cell types

† original question, written from lecture content

15. Which transporter class is primarily responsible for astrocytic uptake of the lactate that astrocytes then export to neurons?

A. GLUT3 glucose transporter
B. EAAT-2 glutamate transporter
C. Monocarboxylate transporter, MCT1/4
D. Na^+/K^+-ATPase pump protein
E. SNAT amino acid transporter

† original question, written from lecture content

16. Which glucose transporter is characteristic of astrocytes, versus the one preferentially expressed in neurons?

A. Astrocyte: GLUT1; Neuron: GLUT3
B. Astrocyte: GLUT3; Neuron: GLUT1
C. Both use GLUT4 exclusively
D. Astrocyte: GLUT2; Neuron: GLUT5
E. Neither cell type expresses glucose transporters

† original question, written from lecture content

17. The pentose phosphate pathway (PPP), activated in neurons especially under high glucose availability, primarily generates which product important for antioxidant defense?

A. ATP directly from the pathway
B. NADPH, regenerating glutathione (GSH)
C. Lactate as the end product
D. Glycogen for storage
E. Acetyl-CoA for the TCA cycle

† original question, written from lecture content

18. Which micronutrient deficiency is classically associated with Wernicke-Korsakoff syndrome, disrupting neuronal energy metabolism?

A. Vitamin B1, thiamine deficiency
B. Vitamin C deficiency
C. Vitamin B12 deficiency only
D. Folate deficiency only
E. Vitamin D deficiency

† original question, written from lecture content

Lectures 5–6 — Embryonic Development of the Nervous System / Development of the Eye (5 questions)

19. Which embryonic structure gives rise to the pineal gland?

A. Roof of the diencephalon (epithalamus)
B. Floor of the mesencephalon region
C. Telencephalon tissue directly
D. Roof of the myelencephalon region
E. Neural crest cell migration

20. Which functional component is characteristic of neurons in the hypoglossal nucleus, located at the level of the myelencephalon (medulla)?

A. GSA (general somatic afferent)
B. SVA (special visceral afferent)
C. GVA (general visceral afferent)
D. SVE (special visceral efferent)
E. GSE (general somatic efferent)

21. Which structure does the optic cup differentiate into?

A. Becomes the lens tissue
B. Becomes the corneal epithelium
C. Becomes retina and iris/ciliary epithelium
D. Becomes vitreous humor only
E. Becomes extraocular muscles

22. Failure of the choroid (retinal) fissure to fuse normally results in which congenital anomaly?

A. Coloboma, a gap in the iris or retina
B. Cataract formation
C. Anophthalmia, absent eye
D. Strabismus, misaligned eyes
E. Glaucoma, raised pressure

23. The spinal trigeminal nucleus (carrying facial pain/temperature) is derived from which embryonic structure?

A. Telencephalon tissue
B. Alar plate, the sensory column
C. Basal plate, the motor column
D. Neural crest cells exclusively
E. Notochord tissue nearby

Lecture 7 — Spinal Cord & Blood Supply (5 questions)

24. A 53-year-old man is hospitalized for acute spinal cord infarction (anterior spinal artery territory). Which sensory modality is characteristically SPARED?

A. Pain sensation specifically
B. Temperature sensation specifically
C. Vibration and proprioception, dorsal column
D. Crude touch sensation
E. Motor function specifically

25. Which of the following is NOT part of the central nervous system?

A. Spinal cord tissue
B. Cerebellum tissue
C. Dorsal root ganglion, a PNS structure
D. Brainstem tissue
E. Cerebral cortex tissue

26. If the anterior spinal artery is occluded, which structure/function is spared, consistent with its blood supply territory?

A. Corticospinal tract function spared
B. Pain and temperature sensation spared
C. Dorsal column function, separate supply
D. Anterior horn motor neurons spared
E. Lateral spinothalamic tract spared

27. Which finding is a primary MOTOR symptom of a complete spinal cord lesion (transection) below the level of injury?

A. Loss of pain sensation only
B. Spastic paralysis after initial flaccid shock
C. Loss of vibration sense only
D. Autonomic dysreflexia exclusively
E. Loss of smell sensation

28. Which structure extends from the pia mater to the dura mater along the lateral aspect of the spinal cord, anchoring it and separating dorsal (posterior) from ventral (anterior) spinal nerve roots?

A. Denticulate ligament
B. Filum terminale
C. Ligamentum flavum
D. Arachnoid trabeculae
E. Falx cerebri

Lecture 8 — PNS: Cranial & Spinal Nerves, Nerve Plexuses (3 questions)

29. Which functional component is NOT found in the facial nerve (CN VII)?

A. SVE, motor to facial expression muscles
B. GVE, parasympathetic to glands
C. SVA, taste from anterior tongue
D. GSA, sensation from external ear
E. GVA, from thoracic/abdominal viscera

30. Which spinal nerves are NOT included in a somatic nerve plexus?

A. C1-C4, cervical plexus region
B. T2-T12, run as intercostal nerves
C. C5-T1, brachial plexus region
D. L1-S4, lumbosacral plexus region
E. All spinal nerves form a plexus

31. A 41-year-old woman has recurring episodes of sharp, electric-shock-like facial pain triggered by chewing and touching her cheek. Which nerve is most likely involved?

A. Facial nerve, Bell's palsy pattern
B. Trigeminal nerve, neuralgia pattern
C. Glossopharyngeal nerve pattern
D. Vagus nerve pattern
E. Accessory nerve pattern

Lectures 9–10 — Brainstem: External & Internal Structures / Reticular Formation (10 questions)

32. Which tract’s second-order neurons cross the midline at the level of the medulla oblongata (rather than immediately upon entering the spinal cord)?

A. Lateral spinothalamic tract (crosses within 1-2 spinal segments)
B. Dorsal column-medial lemniscus pathway (crosses as internal arcuate fibers in the caudal medulla)
C. Anterior spinothalamic tract
D. Corticospinal tract (crosses at the medulla too, but as a first-order/UMN axon, not a sensory 2nd-order neuron)
E. None of the sensory pathways cross in the medulla

33. Damage to which major component of the Ascending Reticular Activating System (ARAS) most directly impairs consciousness?

A. Cerebellum tissue damage
B. Spinal cord below C4 damage
C. Peripheral nerve damage
D. Pons and midbrain reticular formation
E. Basal ganglia damage exclusively

34. What is the major anatomical site whose bilateral damage produces coma via ARAS disruption?

A. Cerebral cortex alone, unilaterally
B. Pons and midbrain reticular formation
C. Cerebellar vermis tissue
D. Spinal cord tissue
E. Peripheral autonomic ganglia

35. A patient presents with altered consciousness, irregular breathing patterns, and decorticate/decerebrate posturing. This clinical picture most directly suggests dysfunction at which level?

A. Peripheral nerve damage only
B. Brainstem reticular formation dysfunction
C. Isolated muscle disease
D. Isolated dorsal root ganglion damage
E. Isolated retinal disease

36. Which cranial nerve nuclei are found at the level of the pons (not the medulla or midbrain)?

A. CN III, IV
B. CN V, VI, VII, VIII
C. CN IX, X, XI, XII
D. CN I, II
E. None; the pons has no cranial nerve nuclei

† original question, written from lecture content

37. Which structure, if damaged bilaterally in the pons, classically produces 'locked-in syndrome' (preserved consciousness and vertical eye movement, but quadriplegia and inability to speak)?

A. Bilateral ventral pons, sparing the tegmentum
B. Bilateral cerebral cortex damage
C. Bilateral cerebellum damage
D. Bilateral thalamus damage
E. Bilateral occipital lobe damage

† original question, written from lecture content

38. Why does a lesion of the medial longitudinal fasciculus (MLF) classically cause internuclear ophthalmoplegia (INO)?

A. MLF links abducens to contralateral oculomotor nucleus
B. MLF is purely sensory, no motor role
C. MLF controls only vertical gaze
D. MLF has no cranial nerve connections
E. MLF sits in the spinal cord, unrelated to eyes

† original question, written from lecture content

39. Which best explains why a lesion of one medial longitudinal fasciculus impairs ADDUCTION of the ipsilateral eye specifically, rather than some other eye movement?

A. MLF carries the abducens signal to oculomotor medial rectus
B. MLF has no relationship to eye muscles
C. MLF controls only vertical eye movements
D. MLF is part of the visual pathway, not motor
E. Adduction is controlled by the trochlear nerve

† original question, written from lecture content

40. Which region does the Ascending Reticular Activating System (ARAS) NOT directly project to?

A. Thalamus, intralaminar nuclei
B. Basal forebrain region
C. Hypothalamus region
D. Cerebral cortex, via thalamic relay
E. Cerebellar deep nuclei, primary target

41. What is the general term for the diffuse network of interconnected neurons running through the core of the brainstem, from the medulla to the midbrain?

A. Reticular formation
B. Corticospinal tract
C. Medial lemniscus
D. Cerebellar peduncle
E. Corpus callosum

† original question, written from lecture content

Lecture 11 — Special Sense Organs & Its Development (8 questions)

42. Cochlear hair cells are innervated by bipolar neurons whose cell bodies reside in which structure?

A. Geniculate ganglion cell bodies
B. Spiral ganglion, within the modiolus
C. Vestibular ganglion cell bodies
D. Petrosal ganglion cell bodies
E. Dorsal root ganglion cell bodies

† original question, written from lecture content

43. Inner hair cells make up what proportion of the afferent fibers of the cochlear nerve, despite being far fewer in number than outer hair cells?

A. About 5%
B. About 50%
C. About 95%
D. 0%, they have no afferent innervation
E. Exactly equal to outer hair cells

† original question, written from lecture content

44. Stereocilia of vestibular/cochlear hair cells are anchored where, and how do they relate to the kinocilium in mechanotransduction?

A. Embedded in a membrane vs kinocilium
B. No relationship to overlying structures
C. Located inside the cell body only
D. Always cause hyperpolarization only
E. Kinocilium absent in mature hair cells

† original question, written from lecture content

45. The three semicircular canals detect which type of head movement?

A. Linear acceleration only, gravity and motion
B. Angular acceleration, each canal a different plane
C. Sound frequency detection only
D. Static head position only, no dynamic sensing
E. Taste changes detection

† original question, written from lecture content

46. Which histological feature distinguishes the utricle and saccule (otolith organs) from the semicircular canal ampullae?

A. Otoconia give maculae mass, unlike crista
B. Identical histology, no real differences
C. Only the ampullae contain hair cells
D. Only the maculae contain hair cells
E. The utricle lacks sensory epithelium

† original question, written from lecture content

47. Which structure supports the organ of Corti and is set into vibration by traveling sound waves, with different regions tuned to different frequencies (tonotopic organization)?

A. Tectorial membrane
B. Basilar membrane
C. Reissner's membrane
D. Stria vascularis
E. Round window membrane

† original question, written from lecture content

48. Presbycusis (age-related hearing loss) classically affects which frequency range first, and which cochlear structure is most vulnerable?

A. Low frequencies first, cochlear apex affected
B. High frequencies first, cochlear base affected
C. All frequencies equally affected
D. Only middle ear ossicles affected
E. Only the vestibular system affected

† original question, written from lecture content

49. Which structure transmits sound vibrations from the tympanic membrane to the oval window, providing mechanical impedance matching between air and cochlear fluid?

A. The three ossicles, acting as a lever system
B. The Eustachian tube structure
C. The round window membrane
D. The semicircular canals
E. The vestibule alone

† original question, written from lecture content

Lectures 12–13 — Somatosensory Pathways / Taste Sensation (8 questions)

50. Which structure is responsible for relaying UNCONSCIOUS proprioceptive information from the lower limb (via the dorsal spinocerebellar tract) toward the cerebellum, synapsing first at Clarke's nucleus (C8-L2/3)?

A. DRG neurons project directly, no relay
B. First-order neuron to Clarke's nucleus to cerebellum
C. Thalamus relays to cerebellum
D. Directly via the corticospinal tract
E. Via the spinothalamic tract

51. What structure is responsible for unconscious proprioceptive input from the SACRAL spinal cord levels (S2-S4), given that Clarke's nucleus does not extend this far caudally?

A. Direct dorsal column ascent, no relay
B. Sacral afferents use an alternate relay route
C. No route to the cerebellum exists at all
D. Travel exclusively via spinothalamic tract
E. Bypass the spinal cord entirely

52. Which pathway carries CONSCIOUS proprioception, fine touch, and vibration sense, as opposed to the unconscious proprioceptive spinocerebellar pathways?

A. Lateral spinothalamic tract pathway
B. Dorsal column-medial lemniscus pathway
C. Dorsal spinocerebellar tract pathway
D. Ventral spinocerebellar tract pathway
E. Anterior spinothalamic tract pathway

† original question, written from lecture content

53. A lesion of the dorsal columns bilaterally (e.g., from vitamin B12 deficiency, subacute combined degeneration) would be expected to cause which finding?

A. Loss of pain with normal proprioception
B. Sensory ataxia, positive Romberg, pain spared
C. Isolated motor weakness, no sensory loss
D. Isolated loss of hearing only
E. Isolated Horner syndrome only

† original question, written from lecture content

54. Which spinal cord tract carries fine touch, vibration, and conscious proprioception from the LOWER limb specifically, ascending in the most medial part of the dorsal columns?

A. Fasciculus cuneatus, upper limb pathway
B. Fasciculus gracilis, lower limb pathway
C. Lateral spinothalamic tract pathway
D. Anterior spinothalamic tract pathway
E. Rubrospinal tract pathway

† original question, written from lecture content

55. An unpleasant or distorted taste sensation is medically termed:

A. Dysphagia
B. Dysphonia
C. Dysgeusia
D. Dysarthria
E. Dysmetria

56. When stereocilia of a taste/hair cell bend TOWARD the kinocilium, what is the physiological result?

A. Hyperpolarization occurs
B. K+ channels close permanently
C. Depolarization occurs
D. Ca2+ efflux increases
E. Action potential frequency decreases

57. In the Gate-Control theory of pain, activated inhibitory spinal interneurons release enkephalins that act by:

A. Blocking Na+ channels presynaptically
B. Opening presynaptic Ca2+ channels
C. Reducing Ca2+ influx and substance P release
D. Stimulating NMDA receptors directly
E. Increasing glutamate release

Lecture 14 — Sensory Receptor Transduction & Neurotransmitters (10 questions)

58. What is the term for the time delay between an action potential arriving at a presynaptic terminal and its effect on the postsynaptic membrane?

A. Synaptic delay
B. Synaptic fatigue

59. Which of the following receptors is ionotropic (directly gates an ion channel upon ligand binding)?

A. AMPA receptor
B. mGluR6
C. GABA-B receptor
D. 5-HT1 receptor

60. What is the mechanism of fluoxetine?

A. Serotonin reuptake inhibitor, blocks SERT
B. Blocks D2 dopamine receptor
C. Blocks central 5-HT3 receptors

61. What is the mechanism of the antiemetic drug ondansetron?

A. Serotonin reuptake inhibitor mechanism
B. Blocks D2 dopamine receptor
C. Blocks peripheral and central 5-HT3 receptors

62. What is the mechanism of haloperidol?

A. Selective serotonin reuptake inhibitor
B. Blocks D2 dopamine receptor
C. Blocks peripheral and central 5-HT3 receptors

63. A patient is splashed with hot cooking oil. Which channel is the primary receptor for NOXIOUS heat in this scenario?

A. TRPV2, very high heat threshold channel
B. TRPV4 channel
C. TRPM8, a cold receptor, wrong category

64. A woman presents with fatigue, social withdrawal, and depressed mood. Which neurotransmitter system is most likely implicated?

A. Dopamine
B. Glutamate
C. Serotonin
D. Acetylcholine
E. Epinephrine

65. Anosmia and ageusia (loss of smell and taste) can result from downregulation of which receptor, also notable for its role as the entry receptor for SARS-CoV-2?

A. Dopamine D2 receptor downregulation
B. Muscarinic acetylcholine receptor
C. ACE2, angiotensin-converting enzyme 2
D. 5-HT3 receptor downregulation
E. NMDA receptor downregulation

66. Which neurotransmitter-transporter pairing is correct?

A. VGAT and GABA, correct vesicular pair
B. SERT and dopamine, mismatched pair
C. DAT and acetylcholine, mismatched pair
D. EAATs and norepinephrine, mismatched
E. Glutaminase and glutamine, mismatched

67. A patient overdoses on a drug that directly opens GABA-A chloride channels even in the complete absence of GABA, causing profound, potentially fatal respiratory depression. Which drug class does this describe, and why is it more dangerous than benzodiazepines in overdose?

A. Benzodiazepines, need GABA present
B. Barbiturates, can gate the channel without GABA
C. SSRIs, no GABA-A activity at all
D. Opioids, unrelated receptor system
E. Antipsychotics, block dopamine not GABA

† original question, written from lecture content

Lectures 15–16 — Visual System / Auditory & Vestibular Systems (10 questions)

68. Which correctly arranges retinal layers from the rod/cone (photoreceptor) layer toward the optic nerve fiber layer?

A. Rods/cones → Inner nuclear → Outer nuclear → Ganglion cell → Optic nerve fiber
B. Rods/cones → Outer nuclear → Inner plexiform → Inner nuclear → Optic nerve fiber
C. Rods/cones → Outer nuclear → Inner nuclear → Ganglion cell → Optic nerve fiber
D. Rods/cones → Ganglion cell → Inner nuclear → Optic nerve fiber

69. A patient has a left pupil that does not constrict and is larger than the right, with associated ptosis and 'down and out' eye deviation. Where is the lesion?

A. Optic tract lesion location
B. Optic nerve lesion location
C. Optic chiasm lesion location
D. CN III, oculomotor nerve lesion
E. Optic radiation lesion location

70. Surface ectoderm gives rise to which eye structure?

A. Lens
B. Iris
C. Corneal endothelium
D. Fibrous tunic (sclera) connective tissue
E. Retina

71. Which specialization of the fovea centralis increases visual acuity there?

A. Abundant blood vessels overlying it
B. High ratio of photoreceptors to ganglion cells
C. Lateral shift of inner retinal layers, less scatter
D. Increased density of rod cells

72. What structure is damaged to produce a RIGHT homonymous hemianopia?

A. Left optic tract or downstream fibers
B. Optic chiasm lesion
C. Superior left LGB region alone
D. Inferior left LGB region alone

73. What structure detects vertical linear acceleration, as when an elevator starts moving up or down?

A. Superior semicircular canal region
B. Lateral semicircular canal region
C. Posterior semicircular canal region
D. Saccule, vertical acceleration sensor
E. Utricle structure nearby

74. What structure primarily detects tilting the head UP (vertical linear head position change)?

A. Utricle
B. Saccule
C. Horizontal semicircular canal
D. Superior semicircular canal
E. Cochlea

75. What structure is primarily activated by nodding the head forward (in the sagittal/vertical rotational plane)?

A. Utricle structure
B. Saccule structure
C. Horizontal semicircular canal
D. Superior semicircular canal, sagittal plane
E. Cochlea structure

76. When you extend your head backward (posterior tilt), which vestibular receptor is primarily stimulated?

A. Superior semicircular canal
B. Lateral semicircular canal
C. Posterior semicircular canal
D. Saccule
E. Utricle

77. Where is the vomiting center located?

A. Vestibular nuclei
B. Nucleus ambiguus
C. Area postrema
D. Superior and inferior salivatory nuclei
E. Nucleus solitarius

Answer Key & Explanations

Lectures 1–16 · 77 Questions

Lecture 1 — Introduction & Organization of the CNS (5 questions)

Q1. Which statement about gray and white matter distribution is correct?

✓ Correct: B. In the cerebrum, gray matter forms the cortex and white matter lies deep to it

Cerebral/cerebellar cortex = gray matter (cell bodies) on the outside; the underlying white matter is myelinated axon tracts connecting cortical areas.

✗ A is wrong: This is backwards for the spinal cord — the spinal cord actually has gray matter centrally (the classic 'butterfly') surrounded by white matter tracts peripherally, the opposite arrangement to the cerebrum.

> *TIP: Spinal cord and cerebrum have OPPOSITE gray/white arrangements — a favorite flip-the-diagram trick question.*

Q2. A newborn is found to have a portion of spinal cord and meninges protruding through a vertebral defect, covered only by a thin membrane....

✓ Correct: C. Myelomeningocele

Myelomeningocele = neural tissue (spinal cord/roots) PLUS meninges herniate through the vertebral defect — the most severe open form of spina bifida.

✗ B is wrong: A meningocele herniates meninges and CSF only — no neural tissue — so it usually has milder or no neurological deficit.

✗ A is wrong: Spina bifida occulta is the mildest form: just a bony vertebral arch defect with skin intact over it, often an incidental finding.

> *TIP: Rank spina bifida severity: occulta (bone only, hidden) < meningocele (meninges herniate) < myelomeningocele (meninges + cord herniate, worst).*

Q3. Which cranial nerve numbers correspond to nerves that are purely sensory?

✓ Correct: B. I, II, VIII

CN I (olfactory), CN II (optic), and CN VIII (vestibulocochlear) carry only special sensory information — no motor fibers.

✗ A is wrong: III, IV, and VI are the purely MOTOR nerves (oculomotor, trochlear, abducens) — the opposite category from what's being asked.

> *TIP: Group the 12 CNs into 3s: Sensory only (I, II, VIII), Motor only (III, IV, VI, XI, XII), Mixed (V, VII, IX, X).*

Q4. Dorsal root ganglia contain the cell bodies of which type of neuron?

✓ Correct: C. First-order primary sensory neurons

DRG house the pseudounipolar cell bodies of primary afferent (first-order sensory) neurons carrying input from the periphery into the spinal cord.

✗ A is wrong: Lower motor neuron cell bodies live in the ventral horn of the spinal cord gray matter, not in a peripheral ganglion.

Q5. A patient has a lesion that damages both anterior and posterior spinal nerve roots equally at one level. Which functional deficit...

✓ Correct: C. Combined sensory and motor loss

The posterior (dorsal) root is sensory and the anterior (ventral) root is motor; damaging both together at a spinal nerve level knocks out both modalities in that dermatome/myotome.

✗ A is wrong: This only accounts for the dorsal root — ignoring the ventral (motor) root damage described in the stem.

> *TIP: Dorsal = sensory (Dorsal = Down = sensory comes down from the periphery... or just remember 'Bell-Magendie law': dorsal root sensory, ventral root motor.*

Lecture 2 — Neurohistology (Neuron & Neuroglia) (3 questions)

Q6. If the tract carrying FAST, sharp, well-localized pain is damaged, which fiber type is responsible for that lost sensation?

✓ Correct: D. Neo-spinothalamic tract, A-δ fibers

Fast, sharp, well-localized ('first') pain is carried by thinly myelinated A-δ fibers via the neospinothalamic tract, which projects rather directly to VPL and then somatosensory cortex — hence good localization.

✗ B is wrong: C fibers carry SLOW, dull, poorly-localized ('second') pain via the paleospinothalamic tract — the opposite pain quality from what's described.

> *TIP: Fast/sharp/localized = A-δ = neospinothalamic. Slow/dull/burning/poorly localized = C fiber = paleospinothalamic (more diffuse projections including to reticular formation/limbic system, which is why it's harder to pinpoint and more 'emotionally' unpleasant).*

Q7. Which of the following is NOT a component of the blood-brain barrier?

✓ Correct: A. Oligodendrocyte

The BBB's neurovascular unit is built from capillary endothelial cells (tight junctions), the basement membrane, pericytes, and astrocytic end-feet — oligodendrocytes play no structural role in it.

✗ C is wrong: Pericytes are a genuine, often-forgotten component that helps regulate BBB permeability and capillary blood flow — students frequently (wrongly) exclude this one instead.

> *TIP: BBB team: Endothelium (tight junctions, the main seal) + Basement membrane + Pericytes + Astrocyte end-feet. Oligodendrocytes are not invited to this party — they're busy myelinating.*

Q8. Which cell type is the resident immune cell of the CNS, analogous to macrophages elsewhere in the body?

✓ Correct: A. Microglia

Microglia are mesoderm/yolk-sac-derived resident CNS immune cells that phagocytose debris and pathogens and mediate neuroinflammation.

✗ D is wrong: Astrocytes support the BBB, metabolism, and repair (glial scarring) but are not the primary phagocytic immune surveillance cell — that's specifically microglia's job.

Lecture 3 — Properties of Neuron: RMP, Action Potential, Conduction (5 questions)

Q9. Which of the following correctly describes when Na+/K+-ATPase activity increases the most?

✓ Correct: A. During high neuronal firing, restoring gradients

Every action potential lets Na+ in and K+ out; the pump has to work harder (using more ATP) proportional to how much firing has occurred, to restore the resting gradients.

Q10. When depolarizations from multiple simultaneous synaptic inputs on different parts of a neuron add together, this is called:

✓ Correct: B. Spatial summation

Spatial summation = combining inputs arriving at different locations on the neuron at roughly the same time; temporal summation is combining repeated inputs at the SAME synapse arriving in rapid succession.

✗ A is wrong: Temporal summation is the time-based version (same synapse, rapid-fire inputs) — the question specifically describes different synapses/locations, which is spatial.

> *TIP: SPATIAL = different SPACE (different synapses) at the same time. TEMPORAL = same synapse, different TIME (rapid repeats).*

Q11. All of the following are true of chemical synapses EXCEPT:

✓ Correct: D. Transmission is instantaneous with essentially zero delay

Chemical synapses always have a measurable synaptic delay (vesicle fusion, diffusion, receptor binding take real time) — instantaneous, zero-delay transmission describes ELECTRICAL synapses (gap junctions), not chemical ones.

> *TIP: Electrical synapses (gap junctions) = fast, bidirectional, no delay, no amplification, but also no plasticity. Chemical synapses = slower, unidirectional, delayed, but capable of amplification and plasticity (learning/memory). Know both columns of this comparison table.*

Q12. Which statement about graded potentials is INCORRECT?

✓ Correct: D. They obey the all-or-none law like action potentials

Graded potentials are explicitly NOT all-or-none — their amplitude scales with stimulus intensity. The all-or-none law is the signature property of the action potential, the opposite category.

Q13. During the absolute refractory period, why can no new action potential be generated no matter how strong the stimulus?

✓ Correct: B. Voltage-gated Na+ channels are inactivated and cannot reopen until the membrane repolarizes

Na+ channels have an inactivation gate that closes shortly after opening and won't reset (become available again) until the membrane has repolarized back toward resting potential — no stimulus, however strong, can force them open during this state.

✗ C is wrong: That describes the RELATIVE refractory period (a stronger-than-normal stimulus CAN trigger an AP), not the absolute refractory period where triggering a new AP is flatly impossible.

> *TIP: Absolute refractory = impossible, period (Na+ channels inactivated). Relative refractory = possible but needs a bigger stimulus (partially hyperpolarized + some channels still inactivated).*

Lecture 4 — Metabolic Pathways of Neurons & Nervous Tissue (5 questions)

Q14. Why can hyperammonemia impair neuronal function even though ammonia detoxification happens mainly in astrocytes rather than neurons?

✓ Correct: A. Astrocyte dysfunction disrupts neuronal fuel/NT recycling

Neurons and astrocytes are metabolically interdependent (lactate shuttle, glutamate-glutamine cycle); when astrocytes are overwhelmed and swollen from ammonia detoxification, this whole support system for neurons breaks down, indirectly impairing neuronal neurotransmission and energy supply even though neurons themselves aren't doing the detoxifying.

> *TIP: Big-picture takeaway questions like this ('why does X affect Y indirectly') are how a prof tests whether you understand the SYSTEM, not just isolated facts. Always ask yourself 'what depends on what?' when reviewing a pathway.*

Q15. Which transporter class is primarily responsible for astrocytic uptake of the lactate that astrocytes then export to neurons?

✓ Correct: C. Monocarboxylate transporter, MCT1/4

Monocarboxylate transporters (MCT1/4 on astrocytes, MCT2 on neurons) move lactate (and pyruvate/ketone bodies) across cell membranes — the physical conduit of the astrocyte-neuron lactate shuttle.

✗ A is wrong: GLUT3 is a neuronal GLUCOSE transporter, not a lactate transporter — glucose and lactate use entirely different transporter families.

Q16. Which glucose transporter is characteristic of astrocytes, versus the one preferentially expressed in neurons?

✓ Correct: A. Astrocyte: GLUT1; Neuron: GLUT3

Astrocytes preferentially express GLUT1 (lower affinity, matches their role as the glucose-sensing/glycolytic 'first responder'); neurons preferentially express GLUT3 (higher affinity, ensures neurons can still pull in glucose even at low extracellular concentrations).

TIP: Higher-affinity transporter goes to the cell type that absolutely cannot afford to run out of substrate — neurons (GLUT3) win the 'first dibs on glucose' contest over astrocytes (GLUT1).

Q17. The pentose phosphate pathway (PPP), activated in neurons especially under high glucose availability, primarily generates which product...

✓ Correct: B. NADPH, regenerating glutathione (GSH)

The PPP's main output relevant here is NADPH, the essential reducing cofactor for regenerating glutathione (GSSG → GSH) and recycling ascorbate — both central to neutralizing oxidative stress, to which neurons are especially vulnerable given their high metabolic rate.

Q18. Which micronutrient deficiency is classically associated with Wernicke-Korsakoff syndrome, disrupting neuronal energy metabolism?

✓ Correct: A. Vitamin B1, thiamine deficiency

Thiamine is a required cofactor for pyruvate dehydrogenase and α-ketoglutarate dehydrogenase (both key TCA cycle enzymes); its deficiency (classically in chronic alcoholism) starves neurons of ATP production and causes the classic triad of confusion, ataxia, and ophthalmoplegia.

TIP: Thiamine deficiency = TCA cycle bottleneck (can't run pyruvate/α-KG dehydrogenase efficiently) = ATP crisis in the most metabolically-demanding brain regions first (mammillary bodies, thalamus) — that's why Wernicke's classically involves those structures.

Lectures 5–6 — Embryonic Development of the Nervous System / Development of the Eye (5 questions)

Q19. Which embryonic structure gives rise to the pineal gland?

✓ Correct: A. Roof of the diencephalon (epithalamus)

The pineal gland develops as an evagination of the diencephalic roof plate (epithalamus) — consistent with it being a genuine CNS/neuroectodermal structure.

Q20. Which functional component is characteristic of neurons in the hypoglossal nucleus, located at the level of the myelencephalon (medulla)?

✓ Correct: E. GSE (general somatic efferent)

The hypoglossal nerve (CN XII) is a pure motor nerve to tongue musculature (derived from occipital somites) — classified functionally as GSE, the same category as the somatic motor nuclei of III, IV, VI.

✗ D is wrong: SVE (special visceral efferent) is the category for branchiomeric muscle motor nuclei (e.g., facial, trigeminal motor, nucleus ambiguus) — tongue muscles are NOT branchial arch derivatives, so hypoglossal is GSE, not SVE.

TIP: CN XII (hypoglossal) and CN III/IV/VI (eye movers) are the only pure GSE cranial nerve nuclei — because tongue and extraocular muscles both derive from somites/somitomeres, not pharyngeal arches.

Q21. Which structure does the optic cup differentiate into?

✓ Correct: C. Becomes retina and iris/ciliary epithelium

The two layers of the optic cup become the neural retina (inner layer) and retinal pigment epithelium (outer layer), with the cup's rim extending forward to form the epithelial layers of the iris and ciliary body.

Q22. Failure of the choroid (retinal) fissure to fuse normally results in which congenital anomaly?

✓ Correct: A. Coloboma, a gap in the iris or retina

A coloboma is a structural defect (notch/gap, classically in the inferior iris/retina/choroid/optic nerve) caused by incomplete fusion of the edges of the choroid fissure during the 5th-7th weeks.

TIP: Colobomas are classically INFERIOR because the choroid fissure runs along the ventral/inferior aspect of the developing optic cup/stalk — if you see 'notch in the iris/retina,' think coloboma, and remember its typical location.

Q23. The spinal trigeminal nucleus (carrying facial pain/temperature) is derived from which embryonic structure?

✓ Correct: B. Alar plate, the sensory column

Sensory brainstem nuclei (including the spinal trigeminal nucleus) derive from the alar plate — the dorsal, sensory-associated portion of the developing neural tube; motor nuclei derive from the ventral basal plate.

✗ C is wrong: The basal plate gives rise to MOTOR nuclei — the sulcus limitans is the developmental landmark separating dorsal alar (sensory) from ventral basal (motor), and this pattern holds throughout the brainstem and spinal cord.

> *TIP: Alar = Afferent/sensory (dorsal). Basal = Basically motor (ventral). This alar/basal split, established early in neural tube development, explains the entire dorsal-sensory/ventral-motor organization you see throughout the spinal cord and brainstem.*

Lecture 7 — Spinal Cord & Blood Supply (5 questions)

Q24. A 53-year-old man is hospitalized for acute spinal cord infarction (anterior spinal artery territory). Which sensory modality is...

✓ Correct: C. Vibration and proprioception, dorsal column

Anterior spinal artery syndrome spares the dorsal columns (supplied instead by the paired posterior spinal arteries), producing a classic dissociated pattern: loss of motor function, pain, and temperature, with PRESERVED vibration/proprioception.

> *TIP: Anterior spinal artery syndrome = motor + pain/temp LOST, proprioception/vibration SPARED (dorsal columns have their own separate posterior spinal artery supply). This is the mirror image of the posterior spinal artery syndrome pattern.*

Q25. Which of the following is NOT part of the central nervous system?

✓ Correct: C. Dorsal root ganglion, a PNS structure

The dorsal root ganglion sits just outside the spinal cord (in or near the intervertebral foramen) and is a PNS structure housing primary sensory neuron cell bodies.

Q26. If the anterior spinal artery is occluded, which structure/function is spared, consistent with its blood supply territory?

✓ Correct: C. Dorsal column function, separate supply

Same key fact restated from the syndrome above: the dorsal columns are spared because they're supplied by the posterior spinal arteries, a separate vascular territory from the anterior spinal artery.

Q27. Which finding is a primary MOTOR symptom of a complete spinal cord lesion (transection) below the level of injury?

✓ Correct: B. Spastic paralysis after initial flaccid shock

Complete cord transection interrupts descending corticospinal (and other motor) tracts, producing an acute phase of flaccid paralysis/areflexia ('spinal shock') that evolves over days-weeks into chronic spastic paralysis with hyperreflexia below the lesion, once spinal reflex circuits recover from the initial shock.

> *TIP: Don't forget the TIME COURSE: immediately after transection = FLACCID (spinal shock), then LATER = SPASTIC. A question describing the acute setting expects flaccid; one describing weeks later expects spastic — a classic timing trap.*

Q28. Which structure extends from the pia mater to the dura mater along the lateral aspect of the spinal cord, anchoring it and separating...

✓ Correct: A. Denticulate ligament

The denticulate ligaments are lateral, tooth-like pial extensions that pierce the arachnoid to attach to the dura, stabilizing the spinal cord within the subarachnoid space and providing a useful surgical landmark between dorsal and ventral roots.

✗ B is wrong: The filum terminale anchors the CAUDAL tip of the spinal cord/conus medullaris to the coccyx — a longitudinal, midline structure, not the lateral tooth-like structures separating the roots.

Lecture 8 — PNS: Cranial & Spinal Nerves, Nerve Plexuses (3 questions)

Q29. Which functional component is NOT found in the facial nerve (CN VII)?

✓ Correct: E. GVA, from thoracic/abdominal viscera

The facial nerve is a genuinely mixed nerve with SVE, GVE, SVA, and a small GSA component (external ear) — but it does NOT carry GVA fibers from thoracic/abdominal viscera; that's characteristic of the vagus nerve instead.

> *TIP: CN VII is famous for carrying FOUR different functional components — a favorite 'which one does it NOT have' trick question, since students often assume a nerve either has one component or 'all of them.'*

Q30. Which spinal nerves are NOT included in a somatic nerve plexus?

✓ Correct: B. T2-T12, run as intercostal nerves

Most thoracic spinal nerves (roughly T2-T12) do NOT form a plexus — they run individually as intercostal (and subcostal) nerves, unlike the cervical, brachial, and lumbosacral levels which do interweave into plexuses.

Q31. A 41-year-old woman has recurring episodes of sharp, electric-shock-like facial pain triggered by chewing and touching her cheek. Which...

✓ Correct: B. Trigeminal nerve, neuralgia pattern

This is the classic presentation of trigeminal neuralgia — brief, severe, electric-shock-like facial pain in a trigeminal (V2/V3 most commonly) distribution, often triggered by light touch, chewing, or talking.

Lectures 9–10 — Brainstem: External & Internal Structures / Reticular Formation (10 questions)

Q32. Which tract's second-order neurons cross the midline at the level of the medulla oblongata (rather than immediately upon entering the...

✓ Correct: B. Dorsal column-medial lemniscus pathway (crosses as internal arcuate fibers in the caudal medulla)

Dorsal column axons ascend ipsilaterally all the way up to the caudal medulla, synapse in the gracile/cuneate nuclei, and THEN the second-order neurons cross as internal arcuate fibers to form the contralateral medial lemniscus — a distinctly later crossing point than the spinothalamic tract's early spinal cord crossing.

✗ A is wrong: Spinothalamic tract fibers cross much earlier and lower — within about 1-2 spinal segments of entering the cord via the anterior white commissure — not at the medulla.

> *TIP: This crossing-LEVEL difference (spinothalamic crosses low, in the cord; DCML crosses high, in the medulla) is EXACTLY why Brown-Séquard syndrome produces opposite-side patterns for these two pathways below a cord lesion, but why a medullary lesion can selectively hit DCML crossing fibers alone.*

Q33. Damage to which major component of the Ascending Reticular Activating System (ARAS) most directly impairs consciousness?

✓ Correct: D. Pons and midbrain reticular formation

The ARAS core is the pontomesencephalic (pons + midbrain) reticular formation, which projects to the thalamus and cortex to maintain wakefulness — bilateral damage here (or to its thalamic/cortical targets) is a classic cause of coma.

Q34. What is the major anatomical site whose bilateral damage produces coma via ARAS disruption?

✓ Correct: B. Pons and midbrain reticular formation

Same core concept: bilateral pontomesencephalic reticular formation damage (or bilateral thalamic/diffuse cortical damage) is required to produce coma — UNILATERAL cortical damage alone typically does not cause coma given the bilateral/diffuse nature of ARAS projections.

> *TIP: Coma requires either (1) bilateral brainstem ARAS damage, or (2) bilateral (diffuse) cortical/thalamic damage — a UNILATERAL hemispheric lesion alone, however large, usually does NOT cause coma unless it secondarily compresses the brainstem (e.g., via herniation).*

Q35. A patient presents with altered consciousness, irregular breathing patterns, and decorticate/decerebrate posturing. This clinical...

✓ **Correct: B. Brainstem reticular formation dysfunction**

Altered consciousness plus abnormal respiratory patterns plus abnormal posturing together localize to brainstem/reticular formation dysfunction (often from rising ICP causing herniation) — respiratory pattern and posturing type can even help localize the LEVEL of brainstem compromise (e.g., decorticate = above red nucleus, decerebrate = below it, at/below midbrain).

> *TIP: Decorticate (arms flexed, legs extended) = lesion ABOVE the red nucleus (better prognosis). Decerebrate (arms AND legs extended) = lesion AT OR BELOW the red nucleus, more caudal, worse prognosis. As the lesion progresses caudally, posturing gets worse — a useful serial-exam trend to recognize.*

Q36. Which cranial nerve nuclei are found at the level of the pons (not the medulla or midbrain)?

✓ **Correct: B. CN V, VI, VII, VIII**

CN V (trigeminal motor/main sensory), VI (abducens), VII (facial), and VIII (vestibulocochlear) nuclei are all located in the pons — a useful grouping for localizing pontine lesions by which nerves are affected.

> *TIP: Cranial nerve nuclei by level: Midbrain = III, IV. Pons = V, VI, VII, VIII. Medulla = IX, X, XI, XII (plus V's spinal nucleus extends down here too). Memorizing this ladder lets you localize ANY brainstem lesion just from which cranial nerves are affected.*

Q37. Which structure, if damaged bilaterally in the pons, classically produces 'locked-in syndrome' (preserved consciousness and vertical eye...

✓ **Correct: A. Bilateral ventral pons, sparing the tegmentum**

Locked-in syndrome results from a lesion (classically basilar artery occlusion) confined to the VENTRAL pons, destroying descending motor pathways (corticospinal/corticobulbar) while SPARING the dorsal tegmentum where the ARAS and vertical gaze centers live — hence consciousness and vertical eye movement are preserved even though the patient cannot move or speak.

> *TIP: This is a devastating and famous localization lesson: ventral pons = 'the highway out' (motor tracts passing through). Dorsal pons/tegmentum = 'the control room' (reticular formation, cranial nerve nuclei). Damage the highway but spare the control room, and you get a fully conscious, aware patient trapped in a paralyzed body — locked-in syndrome.*

Q38. Why does a lesion of the medial longitudinal fasciculus (MLF) classically cause internuclear ophthalmoplegia (INO)?

✓ **Correct: A. MLF links abducens to contralateral oculomotor nucleus**

The MLF is the internuclear connection allowing the abducens nucleus (lateral gaze) to simultaneously signal the contralateral oculomotor nucleus's medial rectus subnucleus, so both eyes move together during horizontal conjugate gaze; damaging it (classically in multiple sclerosis) causes impaired adduction of the ipsilateral eye on attempted lateral gaze, with nystagmus in the abducting eye.

> *TIP: INO is a favorite MS localization question: the affected side is named for which eye fails to ADDUCT (not the side of the primary MLF lesion's laterality convention some students misremember) — always work out the pathway from scratch rather than memorizing a shortcut, since the naming convention trips people up.*

Q39. Which best explains why a lesion of one medial longitudinal fasciculus impairs ADDUCTION of the ipsilateral eye specifically, rather...

✓ **Correct: A. MLF carries the abducens signal to oculomotor medial rectus**

The MLF is specifically the relay from the contralateral abducens nucleus (lateral gaze signal) to the ipsilateral oculomotor medial rectus subnucleus — since medial rectus is the adductor muscle, damaging this specific relay selectively impairs adduction on attempted conjugate gaze, sparing other eye movements that don't depend on this particular circuit.

Q40. Which region does the Ascending Reticular Activating System (ARAS) NOT directly project to?

✓ **Correct: E. Cerebellar deep nuclei, primary target**

The ARAS's defining projections are to the thalamus (intralaminar nuclei), basal forebrain, and hypothalamus, which then relay diffusely to the cortex to maintain arousal — the cerebellum is not its primary/defining target pathway.

Q41. What is the general term for the diffuse network of interconnected neurons running through the core of the brainstem, from the medulla...

✓ Correct: A. Reticular formation

The reticular formation is exactly this: a diffuse, interconnected neuronal network extending through the core of the brainstem from the medulla up through the midbrain, involved in arousal, autonomic regulation, and motor/reflex modulation.

Lecture 11 — Special Sense Organs & Its Development (8 questions)

Q42. Cochlear hair cells are innervated by bipolar neurons whose cell bodies reside in which structure?

✓ Correct: B. Spiral ganglion, within the modiolus

The spiral ganglion, coiled within the bony modiolus at the center of the cochlea, contains the bipolar first-order sensory neurons whose peripheral processes contact hair cells and whose central processes form the cochlear nerve.

✗ C is wrong: The vestibular ganglion (Scarpa’s ganglion) serves the vestibular (balance) hair cells of the semicircular canals/utricle/saccule, a separate ganglion from the cochlear spiral ganglion — even though both eventually join CN VIII.

Q43. Inner hair cells make up what proportion of the afferent fibers of the cochlear nerve, despite being far fewer in number than outer hair...

✓ Correct: C. About 95%

Despite inner hair cells being a minority of total hair cells, they receive about 95% of the afferent (sensory) innervation from the cochlear nerve — they are the true sensory transducers sending sound information to the brain; outer hair cells are mainly EFFERENT-controlled amplifiers (cochlear mechanical tuning), not primary sensory transmitters.

✗ A is wrong: 5% is actually the OUTER hair cell share of afferent innervation — the numbers are commonly swapped in student memory since outer hair cells are more numerous overall but carry far less of the actual afferent signal.

> *TIP: Outer hair cells: more numerous, mostly EFFERENT-controlled amplifiers (cochlear amplifier function), only ~5% of afferent fibers. Inner hair cells: fewer in number, but ~95% of afferent fibers — they’re the ones actually 'talking’ to the brain about sound.*

Q44. Stereocilia of vestibular/cochlear hair cells are anchored where, and how do they relate to the kinocilium in mechanotransduction?

✓ Correct: A. Embedded in a membrane vs kinocilium

Stereocilia (graded in height, embedded in the tectorial membrane in the cochlea, or the otolithic/cupular membrane in the vestibular system) bend when the overlying membrane shifts; bending TOWARD the tallest stereocilia/kinocilium opens mechanically-gated K+ channels causing depolarization, while bending AWAY closes them, causing hyperpolarization.

> *TIP: Direction relative to kinocilium is everything: TOWARD kinocilium = depolarization = more neurotransmitter release = more afferent firing. AWAY from kinocilium = hyperpolarization = less firing. This single directional rule underlies hearing, balance (linear and angular acceleration detection), and is tested constantly.*

Q45. The three semicircular canals detect which type of head movement?

✓ Correct: B. Angular acceleration, each canal a different plane

The three semicircular canals (anterior, posterior, lateral/horizontal), each roughly orthogonal to the others, detect ANGULAR (rotational) acceleration in their respective planes via endolymph movement deflecting the cupula.

✗ A is wrong: Linear acceleration (including gravity/head tilt) is detected by the OTOLITH organs (utricle and saccule), not the semicircular canals — a fundamental and frequently tested vestibular distinction.

> *TIP: Semicircular canals = ANGULAR/rotational acceleration (spinning). Utricle + saccule (otolith organs) = LINEAR acceleration + static head position relative to gravity (tilting, elevator start/stop). Two different physical quantities, two different organ types.*

Q46. Which histological feature distinguishes the utricle and saccule (otolith organs) from the semicircular canal ampullae?

✓ Correct: A. Otoconia give maculae mass, unlike crista

Otoconia (calcium carbonate crystals) embedded in the otolithic membrane give the maculae extra inertial mass, allowing them to lag behind during linear acceleration/gravity shifts and deflect the underlying hair cell stereocilia — the ampullary cupula has no such crystals, relying instead on endolymph flow during rotational movement.

Q47. Which structure supports the organ of Corti and is set into vibration by traveling sound waves, with different regions tuned to...

✓ Correct: B. Basilar membrane

The basilar membrane's mechanical properties vary along its length (narrow/stiff at the base = high frequencies; wide/flexible at the apex = low frequencies), creating tonotopic organization — the physical basis of frequency discrimination.

> *TIP: Base of cochlea = narrow, stiff basilar membrane = HIGH frequency. Apex = wide, floppy basilar membrane = LOW frequency. This base-high/apex-low tonotopic map is the anatomical basis for frequency coding and gets tested via both anatomy and clinical (presbycusis pattern) questions.*

Q48. Presbycusis (age-related hearing loss) classically affects which frequency range first, and which cochlear structure is most vulnerable?

✓ Correct: B. High frequencies first, cochlear base affected

Presbycusis classically begins with HIGH-frequency hearing loss, reflecting cumulative damage to outer (and eventually inner) hair cells at the cochlear BASE — the base being tuned to high frequencies and cumulatively exposed to the most mechanical stress over a lifetime.

Q49. Which structure transmits sound vibrations from the tympanic membrane to the oval window, providing mechanical impedance matching...

✓ Correct: A. The three ossicles, acting as a lever system

The ossicular chain (malleus-incus-stapes) provides mechanical impedance matching (via lever action and the area ratio between the larger tympanic membrane and smaller oval window) — without this amplification, most sound energy would simply reflect off the fluid-air interface rather than being transmitted into the cochlea.

Lectures 12–13 — Somatosensory Pathways / Taste Sensation (8 questions)

Q50. Which structure is responsible for relaying UNCONSCIOUS proprioceptive information from the lower limb (via the dorsal spinocerebellar...

✓ Correct: B. First-order neuron to Clarke's nucleus to cerebellum

Unconscious proprioceptive input from the lower limb synapses first onto second-order neurons in Clarke's nucleus (dorsal nucleus, present roughly C8-L2/3), which then form the DORSAL spinocerebellar tract, an uncrossed pathway carrying fine proprioceptive/movement information to the ipsilateral cerebellum for real-time motor coordination.

> *TIP: Clarke's nucleus is present only C8-L2/3 — below L2/3, unconscious lower-limb proprioception instead uses an alternate route (spinal border cells → ventral spinocerebellar tract) since there's no Clarke's nucleus at those lower levels to synapse on directly. This nucleus-only-at-certain-levels fact is a favorite trick.*

Q51. What structure is responsible for unconscious proprioceptive input from the SACRAL spinal cord levels (S2-S4), given that Clarke's...

✓ Correct: B. Sacral afferents use an alternate relay route

Since Clarke's nucleus proper doesn't extend into the sacral cord, unconscious proprioceptive signals from S2-S4 instead ascend and effectively relay via the ventral spinocerebellar tract pathway (using spinal border cells as the functional equivalent further down), ultimately still reaching the cerebellum for lower-limb coordination.

Q52. Which pathway carries CONSCIOUS proprioception, fine touch, and vibration sense, as opposed to the unconscious proprioceptive...

✓ Correct: B. Dorsal column-medial lemniscus pathway

The DCML pathway is the one that reaches conscious awareness via the thalamus (VPL) and primary somatosensory cortex — the spinocerebellar tracts, by contrast, carry proprioceptive information that never reaches consciousness, feeding directly into cerebellar motor coordination circuits instead.

> *TIP: Key distinction: DCML = CONSCIOUS proprioception (you can tell where your limb is). Spinocerebellar tracts = UNCONSCIOUS proprioception (used only by the cerebellum for automatic movement coordination, never reaches your awareness). Same general 'sense,' two totally different destinations and purposes.*

Q53. A lesion of the dorsal columns bilaterally (e.g., from vitamin B12 deficiency, subacute combined degeneration) would be expected to...

✓ Correct: B. Sensory ataxia, positive Romberg, pain spared

Subacute combined degeneration classically affects the dorsal columns (and lateral corticospinal tracts) bilaterally, sparing the spinothalamic tracts — producing the classic dissociated pattern of proprioceptive/vibratory loss with sensory ataxia (positive Romberg), while pain and temperature sensation remain intact.

Q54. Which spinal cord tract carries fine touch, vibration, and conscious proprioception from the LOWER limb specifically, ascending in the...

✓ Correct: B. Fasciculus gracilis, lower limb pathway

The dorsal columns are somatotopically organized: fasciculus gracilis (medial) carries LOWER limb/lower trunk input, while fasciculus cuneatus (lateral, added above T6) carries UPPER limb/upper trunk input — a favorite localization detail.

✗ A is wrong: Fasciculus cuneatus is the UPPER limb pathway, and it's positioned LATERAL (not medial) to the gracile fasciculus — the exact opposite pairing (nerve/position) from what this lower-limb question describes.

> *TIP: Dorsal column somatotopy: Gracile (medial, 'G for legs, Ground-level, lower/medial') = lower body. Cuneatus (lateral, added above T6) = upper body. This medial-leg/lateral-arm organization is preserved all the way up into the medial lemniscus too.*

Q55. An unpleasant or distorted taste sensation is medically termed:

✓ Correct: C. Dysgeusia

Dysgeusia = distorted/unpleasant taste. The other 'dys-' terms all refer to unrelated deficits: dysphagia (swallowing), dysphonia (voice), dysarthria (articulation), dysmetria (cerebellar movement accuracy).

> *TIP: 'Dys-' word bank, don't mix these up: -phagia=swallowing, -phonia=voice/phonation, -geusia=taste, -arthria=speech articulation, -metria=cerebellar movement accuracy, -osmia=smell (anosmia=no smell). Five to seven similar-sounding terms is a classic examiner trap — build your own glossary.*

Q56. When stereocilia of a taste/hair cell bend TOWARD the kinocilium, what is the physiological result?

✓ Correct: C. Depolarization occurs

Bending toward the kinocilium opens mechanically-gated cation channels, causing depolarization and increased neurotransmitter release/afferent firing — the opposite of bending away, which hyperpolarizes the cell.

Q57. In the Gate-Control theory of pain, activated inhibitory spinal interneurons release enkephalins that act by:

✓ Correct: C. Reducing Ca2+ influx and substance P release

Enkephalins act at opioid receptors to REDUCE presynaptic Ca^{2+} influx (less neurotransmitter/substance P release from the primary nociceptive afferent) AND increase postsynaptic K^+ conductance (hyperpolarizing the second-order pain neuron) — a dual pre- and postsynaptic inhibitory mechanism dampening the ascending pain signal.

> *TIP: Gate-control full circuit: large-diameter mechanoreceptor (A-β) input → activates inhibitory spinal interneurons → enkephalin release → presynaptic Ca2+↓ AND postsynaptic K+↑ → less substance P, less pain transmission. This is literally why rubbing an injury ('gating' pain with touch input) actually works physiologically.*

Lecture 14 — Sensory Receptor Transduction & Neurotransmitters (10 questions)

Q58. What is the term for the time delay between an action potential arriving at a presynaptic terminal and its effect on the postsynaptic...

✓ Correct: A. Synaptic delay

Synaptic delay (~0.5-1ms) reflects the time needed for Ca2+ entry, vesicle fusion, neurotransmitter diffusion, and receptor binding — the real, measurable time cost of chemical (as opposed to electrical) synaptic transmission.

✗ B is wrong: Synaptic fatigue refers to a DECREASE in synaptic efficacy with sustained/repeated use (neurotransmitter depletion), a completely different phenomenon from the fixed time delay of a single transmission event.

Q59. Which of the following receptors is ionotropic (directly gates an ion channel upon ligand binding)?

✓ Correct: A. AMPA receptor

AMPA receptors are ionotropic glutamate receptors (ligand-gated Na+/K+ channels), providing fast excitatory transmission — the other three listed (mGluR6, GABA-B, 5-HT1) are all metabotropic (G-protein-coupled), acting through slower second-messenger cascades.

> *TIP: Ionotropic = fast, direct ion channel (AMPA, NMDA, GABA-A, glycine, nicotinic ACh, 5-HT3). Metabotropic = slow, G-protein-coupled (mGluRs, GABA-B, muscarinic ACh, most 5-HT subtypes, dopamine, adrenergic). '3' is the magic ionotropic serotonin number to remember: 5-HT3 is the ONE ionotropic serotonin receptor; all other 5-HT subtypes are metabotropic.*

Q60. What is the mechanism of fluoxetine?

✓ Correct: A. Serotonin reuptake inhibitor, blocks SERT

Fluoxetine is a classic SSRI, blocking SERT to increase synaptic serotonin.

Q61. What is the mechanism of the antiemetic drug ondansetron?

✓ Correct: C. Blocks peripheral and central 5-HT3 receptors

Ondansetron is a 5-HT3 receptor antagonist, blocking serotonin's action at this specific ionotropic receptor subtype involved in the vomiting reflex.

Q62. What is the mechanism of haloperidol?

✓ Correct: B. Blocks D2 dopamine receptor

Haloperidol is a typical (first-generation) antipsychotic, blocking D2 dopamine receptors.

> *TIP: Three drugs, three completely different single-target mechanisms tested back to back (fluoxetine=SERT, ondansetron=5-HT3, haloperidol=D2) — a favorite examiner trick is to present these consecutively and swap the mechanisms between them.*

Q63. A patient is splashed with hot cooking oil. Which channel is the primary receptor for NOXIOUS heat in this scenario?

✓ Correct: A. TRPV2, very high heat threshold channel

For truly extreme/noxious heat (like hot oil, well above the ~43°C threshold of TRPV1), TRPV2 is the higher-threshold heat nociceptor channel recruited — distinct from TRPV1's more moderate noxious-heat range.

> *TIP: Thermal TRP channel ladder by temperature: TRPM8 (cool, ~menthol) < TRPV4/TRPV3 (warm) < TRPV1 (hot/noxious, ~43°C+, capsaicin) < TRPV2 (extreme heat, >52°C). The hotter the stimulus, the higher up this ladder you go.*

Q64. A woman presents with fatigue, social withdrawal, and depressed mood. Which neurotransmitter system is most likely implicated?

✓ Correct: C. Serotonin

Same core depression-serotonin link tested again — a very frequently repeated exam fact worth over-learning.

Q65. Anosmia and ageusia (loss of smell and taste) can result from downregulation of which receptor, also notable for its role as the entry...

✓ Correct: C. ACE2, angiotensin-converting enzyme 2

ACE2 is expressed on olfactory support cells (sustentacular cells) and is both the receptor SARS-CoV-2 uses for cellular entry and a plausible mechanistic link to COVID-19-associated anosmia/ageusia — a great example of a basic science fact suddenly becoming extremely clinically relevant.

Q66. Which neurotransmitter-transporter pairing is correct?

✓ Correct: A. VGAT and GABA, correct vesicular pair

VGAT loads GABA into synaptic vesicles — a correctly matched pair. The other four options mismatch a transporter/enzyme with the wrong neurotransmitter (SERT is for serotonin not dopamine; DAT is for dopamine not ACh; EAATs are for glutamate not norepinephrine; glutaminase is an enzyme, not a transporter, and it converts glutamine TO glutamate, it doesn't transport glutamine per se).

Q67. A patient overdoses on a drug that directly opens GABA-A chloride channels even in the complete absence of GABA, causing profound,...

✓ Correct: B. Barbiturates, can gate the channel without GABA

Barbiturates, unlike benzodiazepines, can directly open the GABA-A chloride channel even without GABA bound, at sufficiently high concentrations — this GABA-independent action removes any physiological 'ceiling' on inhibition, making barbiturate overdose far more likely to cause fatal respiratory arrest than benzodiazepine overdose (where the effect is capped by needing GABA to be present).

> *TIP: This 'why is X more dangerous than Y' framing is a hallmark of a well-designed critical-thinking question — always be ready to explain not just the mechanism, but its clinical CONSEQUENCE (here: no ceiling effect = more dangerous in overdose).*

Lectures 15–16 — Visual System / Auditory & Vestibular Systems (10 questions)

Q68. Which correctly arranges retinal layers from the rod/cone (photoreceptor) layer toward the optic nerve fiber layer?

✓ Correct: C. Rods/cones → Outer nuclear → Inner nuclear → Ganglion cell → Optic nerve fiber

Correct order from photoreceptor toward the vitreous/optic nerve: photoreceptor segments → outer nuclear layer (photoreceptor cell bodies) → outer plexiform → inner nuclear layer (bipolar/horizontal/amacrine cell bodies) → inner plexiform → ganglion cell layer → nerve fiber layer.

> *TIP: Remember: light must pass THROUGH most retinal layers before hitting the photoreceptors (the retina is 'inside-out' relative to a camera) — but the SIGNAL then flows the opposite direction, from photoreceptor outward toward the ganglion cells and optic nerve, which is the direction this question tests.*

Q69. A patient has a left pupil that does not constrict and is larger than the right, with associated ptosis and 'down and out' eye...

✓ Correct: D. CN III, oculomotor nerve lesion

Dilated, fixed pupil + ptosis + down-and-out eye = complete CN III palsy — the parasympathetic pupillary fibers on the outside of the nerve are compressed, explaining the pupil involvement.

Q70. Surface ectoderm gives rise to which eye structure?

✓ Correct: A. Lens

The lens is the classic surface ectoderm derivative (via lens placode/vesicle) — the other options are neuroectoderm (iris epithelium, retina) or neural crest/mesoderm (corneal endothelium, sclera) derived.

Q71. Which specialization of the fovea centralis increases visual acuity there?

✓ Correct: C. Lateral shift of inner retinal layers, less scatter

At the fovea, the inner retinal layers (ganglion cells, bipolar cells, etc.) are pushed laterally out of the way, letting light hit the densely-packed cone photoreceptors with minimal obstruction/scatter — a key structural adaptation for maximal visual acuity.

✗ D is wrong: The fovea is actually CONE-dominated (essentially no rods at its very center) — increased ROD density is the opposite of what makes the fovea special; rods dominate the peripheral retina instead, favoring sensitivity over acuity.

> *TIP: Fovea = cones only, best acuity, worst in dim light (no rods = can't see faint stars if you look straight at them — that's why astronomers use averted vision, shifting the image to the rod-rich periphery).*

Q72. What structure is damaged to produce a RIGHT homonymous hemianopia?

✓ Correct: A. Left optic tract or downstream fibers

A right homonymous hemianopia (loss of the right visual field in BOTH eyes) results from a lesion of the LEFT visual pathway AFTER the chiasm (left optic tract, radiation, or occipital cortex) — post-chiasmal lesions always cause a homonymous (same-side) field cut, on the side OPPOSITE the lesion.

> *TIP: Golden rule: PRE-chiasmal lesions (optic nerve) = monocular defects. AT the chiasm = bitemporal (heteronymous). POST-chiasmal (tract, radiation, cortex) = homonymous, on the side of the visual field OPPOSITE to the lesion side. Always work out which 'zone' the lesion is in first.*

Q73. What structure detects vertical linear acceleration, as when an elevator starts moving up or down?

✓ Correct: D. Saccule, vertical acceleration sensor

The saccule's maculae are oriented in a roughly VERTICAL plane, making it the primary detector of vertical linear acceleration (elevator start/stop, jumping) — while the utricle's horizontally-oriented macula detects horizontal linear acceleration instead.

✗ E is wrong: The utricle detects HORIZONTAL linear acceleration (like a car starting to move forward), the perpendicular plane from what this vertical-elevator scenario is testing.

> *TIP: Saccule = VERTICAL linear acceleration (elevator, jumping) — think 'Saccule = Straight up and down.' Utricle = HORIZONTAL linear acceleration (car accelerating forward) — think 'Utricle = yoU're driving forward.'*

Q74. What structure primarily detects tilting the head UP (vertical linear head position change)?

✓ Correct: B. Saccule

Same saccule/vertical-plane logic as the elevator question — tilting the head to change its vertical position relative to gravity is detected by the vertically-oriented saccular macula.

Q75. What structure is primarily activated by nodding the head forward (in the sagittal/vertical rotational plane)?

✓ Correct: D. Superior semicircular canal, sagittal plane

Nodding ('yes' motion) is a rotational movement in the sagittal plane, detected by the superior (anterior) semicircular canal, which is oriented to sense exactly this plane of angular acceleration.

> *TIP: Semicircular canal orientation-to-movement mapping: Horizontal canal = shaking head 'no' (yaw). Anterior/superior canal = nodding 'yes' (pitch). Posterior canal = tilting head toward shoulder (roll). Each canal is tuned to detect rotation specifically in its own anatomical plane.*

Q76. When you extend your head backward (posterior tilt), which vestibular receptor is primarily stimulated?

✓ Correct: A. Superior semicircular canal

Extending the head backward is a rotational (pitch-plane) movement, sensed by the anterior/superior semicircular canal — the same canal responsible for the forward-nodding motion, since it detects rotation in that plane in EITHER direction.

Q77. Where is the vomiting center located?

✓ Correct: C. Area postrema

Area postrema (a circumventricular organ with a leaky BBB) functions as the chemoreceptor trigger zone that detects blood-borne emetogenic substances and directly triggers vomiting.

TIP: Note the connection to vestibular input here too: motion sickness works because vestibular nuclei ALSO project to the vomiting center/area postrema — that's the whole physiological basis of why spinning/motion makes you nauseated.